MUSTAPHA O. SHOMORIN

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with experience designing, building and monitoring end-to-end data pipelines and platforms across clinical research, finance, retail, HR and logistics. Skilled in ETL/ELT with Python and SQL, batch and streaming ingestion, medallion (Bronze/Silver/Gold) architecture, dimensional modelling, orchestration with Airflow and dbt, automated data quality validation, and pipeline observability with Prometheus and Grafana. Experienced in deploying data and ML services through REST APIs and Docker on AWS, Azure and GCP, and in delivering BI dashboards in Power BI, Tableau and Streamlit. Reduced manual reporting time by up to 50% through automation. Strong communicator who translates business needs into reliable data solutions and mentors junior analysts. MSc Financial Engineering in progress (WorldQuant University, 2027); BSc Physics, First Class Honours.

TECHNICAL SKILLS

Data Engineering: ETL/ELT pipelines, batch and streaming ingestion, Apache Airflow, dbt, Apache Spark, Redpanda (Kafka-compatible streaming), medallion (Bronze/Silver/Gold) architecture, dimensional modelling, retry logic and replay protection

Languages: Python (Pandas, NumPy), SQL, R, SAS, MATLAB

Databases & Storage: DuckDB, PostgreSQL, MySQL, SQLite, MongoDB, object storage

Cloud & DevOps: AWS, Azure, GCP, Docker, Git/GitHub, CI/CD workflows

Data Quality & Observability: Validation frameworks, checks for critical fields, uniqueness, referential integrity and business rules, unit testing of transformations, anomaly detection, Prometheus, Grafana, audit trails

Serving & BI: Streamlit, FastAPI, Flask, REST APIs, Power BI, Tableau, Plotly Dash, Excel (Power Query, VBA, dynamic arrays)

ML & Analytics: Scikit-learn, TensorFlow, Keras, PyTorch, NLP (NLTK, spaCy), time-series forecasting, SHAP, LIME, Statsmodels

Standards & Compliance: CDISC (SDTM/ADaM), EDC systems, HIPAA, GDPR, data governance

PROFESSIONAL EXPERIENCE

ClinFocus, Atlanta, Georgia, USA **Jan 2026 – Jun 2026**
Clinical Data Scientist

- Developed data ingestion and processing pipelines using SQL, Python, R, Spark and cloud platforms (Azure/AWS/GCP).
- Collected, cleaned, integrated and validated clinical and real-world datasets (EHR, EMR, registries, claims, lab systems, wearables).
- Applied CDISC standards (SDTM, ADaM) and maintained documentation and audit trails in line with HIPAA and GDPR requirements.
- Built automated reporting and dashboards (Excel, Power BI, Tableau, Python/R) for enrollment, EDC query and patient-level data review.
- Built predictive models for patient stratification, disease progression and risk scoring, and validated them with appropriate statistical techniques.
- Worked with clinical operations, medical affairs and biostatistics teams on protocol requirements, data review and TFL preparation.

Vigour Technology, Lagos, Nigeria **Jan 2025 – Dec 2025**
Senior Data Scientist & Head of Sales (focus: data engineering, ML deployment and client delivery)

- Designed, built and deployed ETL/ELT pipelines in Python and SQL, ingesting structured and unstructured data from APIs, databases, cloud storage and third-party SaaS platforms.
- Implemented data validation, pipeline reliability monitoring and error handling, plus model monitoring for drift, performance degradation and retraining triggers.
- Deployed models and analytics services through REST APIs, FastAPI and Docker, using AWS, Azure and Airflow for hosting, storage and scheduling.
- Built reusable Python modules and internal packages; set up Git-based workflows with branching, pull requests and CI/CD review for all data projects.
- Documented data schemas, model versions and deployment procedures for internal use and client handover.
- Delivered data and BI solutions (Power BI, Tableau, Plotly Dash) for finance, retail, health and logistics clients; ML-based automation reduced manual reporting time by up to 50%.
- Led client engagements from needs assessment to delivery, and mentored junior data scientists and analysts on coding standards and project execution.
- As Head of Sales, led the sales team to grow the client base using data-driven strategy, forecasting and customer segmentation, and wrote client proposals and solution architectures.

Zidio Development **Jan 2025 – Apr 2025**

Data Science / Data Engineering Intern

- Built automated data pipelines and BI systems, structured datasets and ran data cleaning and quality validation ahead of model training.
- Implemented real-time data monitoring for pipeline health, reliability and anomaly detection; strengthened data governance and security practices.
- Developed multimodal (text, audio, image) emotion-detection models for the Zidio AI-Powered Task Optimizer and built Power BI dashboards to track model performance.

Cognifyz Technologies **Jan 2025 – Mar 2025**

Data Science Intern

- Led an end-to-end restaurant operations analytics project covering performance trends, customer behaviour and operational bottlenecks.
- Assessed data accuracy, completeness, consistency and integrity, and built automated anomaly detection and real-time data quality monitoring.
- Delivered Power BI dashboards and management reports with insights and recommendations.

University of Ibadan, Ibadan, Nigeria **Jan 2024 – Oct 2024**

Data Management Analyst (NYSC)

- Automated data collection, validation and reporting workflows, reducing manual processing errors by 30%.
- Managed large-scale staff and student records with over 99% data accuracy; ran routine data audits for completeness, consistency and governance compliance.
- Built reporting templates and documented data procedures; received an Award of Excellence for teamwork, innovation and operational efficiency.

SELECTED PROJECTS

NorthStar Retail: End-to-End Retail Data Platform *Oct 2026*

- Built an end-to-end data platform with Bronze/Silver/Gold architecture: batch ingestion of a historical retail dataset plus synthetic order and customer events to demonstrate streaming.
- Stack: Python and Redpanda for ingestion and event streaming, DuckDB for analytical storage, dbt for SQL transformations and dimensional modelling, Apache Airflow for orchestration.
- Added retry behaviour, failure notifications, replay protection, automated data quality checks (critical fields, uniqueness, referential integrity, business rules) and unit tests for custom transformations.
- Monitored pipeline status, failures, records processed and data freshness with Prometheus and Grafana; served a five-section Streamlit executive dashboard (revenue, geography, products, customers, delivery operations).

Real-Time Weather Data ETL & Visualization Pipeline *Mar 2025*

- Automated collection, cleaning and storage of live OpenWeatherMap data with scheduled updates, issue tracking and an interactive dashboard.

Zidio AI-Powered Task Optimizer *Mar 2025*

- Built a multimodal emotion-detection system (text, audio, facial expressions) linked to task assignment, with trend tracking and stress/burnout alerts (TensorFlow, PyTorch, Scikit-learn, Streamlit).

EDC Query Management & Data Quality Analytics *Mar 2026*

- Analysed query volume, aging, resolution timelines and site-level query rates; built automated Excel reporting that identified high-risk sites.

Clinical Patient Profile Dashboard *May 2026*

- Automated a consolidated patient-level view from multiple clinical datasets (dynamic arrays, XLOOKUP, LET, FILTER) with instant subject switching from a single input.

Clinical Study Enrollment Analytics Dashboard *Apr 2026*

- Tracked enrollment trends, screen failure and randomization rates, site activation timelines and enrollment velocity to expose recruitment bottlenecks.

Cryptocurrency Price Prediction with SARIMAX *Mar 2025*

- Built a forecasting model using RSI, SMA and EMA indicators with low prediction errors, deployed through a Flask web interface; code shared on GitHub.

Customer Segmentation Model *Feb 2025*

- Clustering-based customer segmentation with a dashboard; contributed to a 15% increase in customer retention.

Sentiment Analysis on Product Reviews *Dec 2024*

- Built a classifier with 85% accuracy and a web dashboard showing sentiment trends over time.

EDUCATION & CERTIFICATIONS

WorldQuant University, USA **Expected Jun 2027**
MSc Financial Engineering (in progress)

WorldQuant University, USA **Oct 2025**
Certificate: Foundation of Financial Engineering

WorldQuant University, USA **Oct 2024**
Certificate Program in Applied Data Science

University of Ilorin, Ilorin, Kwara State, Nigeria **Oct 2023**
BSc Physics (First Class Honours)

AWARDS

- Award of Excellence, University of Ibadan (NYSC): teamwork, innovation and operational efficiency in managing university data systems.